

Modelling Particle Production in Analog Expanding Universes with High-Performance Computing

Nathan Chapman

Department of Computer Science
Department of Physics
Central Washington University

CWU SOURCE 2024



Introduction

- ① Big Bang and rapid expansion $\implies$ could observe particles
- ② Big Bang was 14 billion years ago $\implies$ we can't observe particles
- ③ Condensed matter dynamics match particle dynamics (see Methods)
- ④ Can make condensed matter in the lab $\implies$ can effectively observe Big Bang particles
- ⑤ Lab based condensed matter system = “analog universe”

Related Work

Previous research has been done investigating a variety of different systems

- 1 “Hyper-phononic” quasiparticles and Lorentz Violation
- 2 Analog massive particles
- 3 Analog particles with non-zero quantum spin
- 4 Analog universes with different expansion behavior
- 5 Analog universes with non-zero background velocity
- 6 Experimental realizations (some extremely relevant and recent)

This Work

My focus is on

- ① $(3+1)$ D analog universe with high-performance computing
- ② linear dispersion & phononic quasiparticle
- ③ analog massless particles
- ④ analog spin-0 particles
- ⑤ analog de Sitter expansion
- ⑥ no background velocity

Theory - Speed & Expansion

- Speed of sound in BEC = Speed of light in analog universe
- Light speed decrease & static universe $\iff$ constant light speed & expanding universe
- Sound speed decrease & static gas $\iff$ constant sound speed & expanding gas
- Example: $t = d/v$
 - Distance to grocery store doubles $\implies$ travel time doubles
 - Speed to the grocery store halves $\implies$ travel time doubles

Theory - The Field Equation

- The field equation describes the behavior of
 - a gas of quantum particles made in the lab
 - particles produced just after the Big Bang
- For each 3D wave vector $\mathbf{k}$ such that $||\mathbf{k}|| < k_c$ and $\hbar^2 k_c^2 / 2m = U n_0$
- Universe expansion starts and stops at finite times $\implies \eta_i \leq \eta \leq \eta_f$

The Field Equation

$$\partial_\eta^2 \tilde{\theta}_{\mathbf{k}} - \frac{1}{2} \frac{\dot{a}(\eta)}{a(\eta)} \partial_\eta \tilde{\theta}_{\mathbf{k}} - c_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0 \quad (1a)$$

$$\tilde{\theta}_{\mathbf{k}}(0) = \tilde{\theta}_{\mathbf{k}0} \quad \partial_\eta \tilde{\theta}_{\mathbf{k}}(0) = -\frac{U_0}{\hbar} n_{\mathbf{k}} \quad (1b)$$

Theory - Numerical Integrator

- Energy needs to be conserved $\implies$ use symplectic integrator (McAte5)
- psuedo code for McAte5 algorithm2e
- error calculations

Theory - Computational Implementation

- Parallel - CPU
- Parallel - GPU
- Use supercomputer

Results

Discussion

Conclusion

Acknowledgements

- Dr. Andy Piacsek
- Dr. Szilard VAJDA
- Dr. Micheal Braunstein

Bibliography I

- [1] M. H. Anderson et al. “Observation of Bose-Einstein Condensation in a Dilute Atomic Vapor”. In: *Science* 269.5221 (1995), pp. 198–201. DOI: [10.1126/science.269.5221.198](https://doi.org/10.1126/science.269.5221.198). eprint: <https://www.science.org/doi/pdf/10.1126/science.269.5221.198>. URL: <https://www.science.org/doi/abs/10.1126/science.269.5221.198>.
- [2] Carlos Barceló, S. Liberati, and Matt Visser. “Probing semiclassical analog gravity in Bose-Einstein condensates with widely tunable interactions”. In: *Phys. Rev. A* 68 (5 Nov. 2003), p. 053613. DOI: [10.1103/PhysRevA.68.053613](https://doi.org/10.1103/PhysRevA.68.053613). URL: <https://link.aps.org/doi/10.1103/PhysRevA.68.053613>.

Bibliography II

- [3] Carlos Barceló, Stefano Liberati, and Matt Visser. “Analogue Gravity”. In: *Living Reviews in Relativity* 8.1 (Dec. 2005). DOI: [10.12942/lrr-2005-12](https://doi.org/10.12942/lrr-2005-12). URL: <https://doi.org/10.12942%5C%2Flrr-2005-12>.
- [4] N. D. Birrell and P. C. W. Davies. *Quantum Fields in Curved Space*. Cambridge Monographs on Mathematical Physics. Cambridge University Press, 1982. DOI: [10.1017/CB09780511622632](https://doi.org/10.1017/CB09780511622632).
- [5] S. L. Cornish et al. “Stable ^{85}Rb Bose-Einstein Condensates with Widely Tunable Interactions”. In: *Phys. Rev. Lett.* 85 (9 Aug. 2000), pp. 1795–1798. DOI: [10.1103/PhysRevLett.85.1795](https://link.aps.org/doi/10.1103/PhysRevLett.85.1795). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.85.1795>.

Bibliography III

- [6] Petr O. Fedichev and Uwe R. Fischer. ““Cosmological” quasiparticle production in harmonically trapped superfluid gases”. In: *Phys. Rev. A* 69 (3 Mar. 2004), p. 033602. DOI: [10.1103/PhysRevA.69.033602](https://doi.org/10.1103/PhysRevA.69.033602). URL: <https://link.aps.org/doi/10.1103/PhysRevA.69.033602>.
- [7] Petr O. Fedichev and Uwe R. Fischer. “Gibbons-Hawking Effect in the Sonic de Sitter Space-Time of an Expanding Bose-Einstein-Condensed Gas”. In: *Phys. Rev. Lett.* 91 (24 Dec. 2003), p. 240407. DOI: [10.1103/PhysRevLett.91.240407](https://doi.org/10.1103/PhysRevLett.91.240407). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.91.240407>.

Bibliography IV

- [8] Petr O. Fedichev and Uwe R. Fischer. “Observer dependence for the phonon content of the sound field living on the effective curved space-time background of a Bose-Einstein condensate”. In: *Phys. Rev. D* 69 (6 Mar. 2004), p. 064021. DOI: [10.1103/PhysRevD.69.064021](https://doi.org/10.1103/PhysRevD.69.064021). URL: <https://link.aps.org/doi/10.1103/PhysRevD.69.064021>.
- [9] Stephen A. Fulling. *Aspects of Quantum Field Theory in Curved Spacetime*. London Mathematical Society Student Texts. Cambridge University Press, 1989. DOI: [10.1017/CB09781139172073](https://doi.org/10.1017/CB09781139172073).

Bibliography V

- [10] C W Gardiner, J R Anglin, and T I A Fudge. “The stochastic Gross-Pitaevskii equation”. In: *Journal of Physics B: Atomic, Molecular and Optical Physics* 35.6 (Mar. 2002), p. 1555. DOI: [10.1088/0953-4075/35/6/310](https://doi.org/10.1088/0953-4075/35/6/310). URL: <https://dx.doi.org/10.1088/0953-4075/35/6/310>.
- [11] G. W. Gibbons and S. W. Hawking. “Cosmological event horizons, thermodynamics, and particle creation”. In: *Phys. Rev. D* 15 (10 May 1977), pp. 2738–2751. DOI: [10.1103/PhysRevD.15.2738](https://link.aps.org/doi/10.1103/PhysRevD.15.2738). URL: <https://link.aps.org/doi/10.1103/PhysRevD.15.2738>.

Bibliography VI

- [12] Alan H. Guth. “Inflationary universe: A possible solution to the horizon and flatness problems”. In: *Phys. Rev. D* 23 (2 Jan. 1981), pp. 347–356. DOI: [10.1103/PhysRevD.23.347](https://doi.org/10.1103/PhysRevD.23.347). URL: <https://link.aps.org/doi/10.1103/PhysRevD.23.347>.
- [13] Ted Jacobson. *Introduction to Quantum Fields in Curved Spacetime and the Hawking Effect*. 2004. arXiv: [gr-qc/0308048](https://arxiv.org/abs/gr-qc/0308048) [gr-qc].
- [14] Piyush Jain et al. “Analog model of a Friedmann-Robertson-Walker universe in Bose-Einstein condensates: Application of the classical field method”. In: *Phys. Rev. A* 76 (3 Sept. 2007), p. 033616. DOI: [10.1103/PhysRevA.76.033616](https://doi.org/10.1103/PhysRevA.76.033616). URL: <https://link.aps.org/doi/10.1103/PhysRevA.76.033616>.

Bibliography VII

- [15] A. S. Lapedes. “Bogoliubov transformations, propagators, and the Hawking effect”. In: *Journal of Mathematical Physics* 19.11 (Aug. 2008), pp. 2289–2293. ISSN: 0022-2488. DOI: 10.1063/1.523607. eprint: https://pubs.aip.org/aip/jmp/article-pdf/19/11/2289/8149302/2289_1_online.pdf. URL: <https://doi.org/10.1063/1.523607>.
- [16] Stefano Liberati, Matt Visser, and Silke Weinfurtner. “Analogue quantum gravity phenomenology from a two-component Bose–Einstein condensate”. In: *Classical and Quantum Gravity* 23.9 (Apr. 2006), p. 3129. DOI: 10.1088/0264-9381/23/9/023. URL: <https://dx.doi.org/10.1088/0264-9381/23/9/023>.

Bibliography VIII

- [17] Stefano Liberati, Matt Visser, and Silke Weinfurtner. “Naturalness in an Emergent Analogue Spacetime”. In: *Phys. Rev. Lett.* 96 (15 Apr. 2006), p. 151301. DOI: [10.1103/PhysRevLett.96.151301](https://doi.org/10.1103/PhysRevLett.96.151301). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.96.151301>.
- [18] R I McLachlan and P Atela. “The accuracy of symplectic integrators”. In: *Nonlinearity* 5.2 (Mar. 1992), p. 541. DOI: [10.1088/0951-7715/5/2/011](https://doi.org/10.1088/0951-7715/5/2/011). URL: <https://dx.doi.org/10.1088/0951-7715/5/2/011>.
- [19] J. L. Roberts et al. “Resonant Magnetic Field Control of Elastic Scattering in Cold ^{85}Rb ”. In: *Phys. Rev. Lett.* 81 (23 Dec. 1998), pp. 5109–5112. DOI: [10.1103/PhysRevLett.81.5109](https://doi.org/10.1103/PhysRevLett.81.5109). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.81.5109>.

Bibliography IX

- [20] Ralf Schützhold, Günter Plunien, and Gerhard Soff. “Dielectric Black Hole Analogs”. In: *Phys. Rev. Lett.* 88 (6 Jan. 2002), p. 061101. DOI: [10.1103/PhysRevLett.88.061101](https://doi.org/10.1103/PhysRevLett.88.061101). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.88.061101>.
- [21] Paul J. Steinhardt and Neil Turok. “Cosmic evolution in a cyclic universe”. In: *Phys. Rev. D* 65 (12 May 2002), p. 126003. DOI: [10.1103/PhysRevD.65.126003](https://doi.org/10.1103/PhysRevD.65.126003). URL: <https://link.aps.org/doi/10.1103/PhysRevD.65.126003>.
- [22] Michael Uhlmann, Yan Xu, and Ralf Schützhold. “Aspects of cosmic inflation in expanding Bose–Einstein condensates”. In: *New Journal of Physics* 7.1 (Nov. 2005), p. 248. DOI: [10.1088/1367-2630/7/1/248](https://doi.org/10.1088/1367-2630/7/1/248). URL: <https://dx.doi.org/10.1088/1367-2630/7/1/248>.

Bibliography X

- [23] Matt Visser. “Some general bounds for one-dimensional scattering”. In: *Phys. Rev. A* 59 (1 Jan. 1999), pp. 427–438. DOI: [10.1103/PhysRevA.59.427](https://doi.org/10.1103/PhysRevA.59.427). URL: <https://link.aps.org/doi/10.1103/PhysRevA.59.427>.
- [24] MATT VISSER, CARLOS BARCELÓ, and STEFANO LIBERATI. “ACOUSTICS IN BOSE–EINSTEIN CONDENSATES AS AN EXAMPLE OF BROKEN LORENTZ SYMMETRY”. In: *CPT and Lorentz Symmetry*, pp. 336–340. DOI: [10.1142/9789812778123_0043](https://doi.org/10.1142/9789812778123_0043). eprint: https://www.worldscientific.com/doi/pdf/10.1142/9789812778123_0043. URL: https://www.worldscientific.com/doi/abs/10.1142/9789812778123_0043.

Bibliography XI

- [25] Matt Visser and Silke Weinfurtner. “Massive Klein-Gordon equation from a Bose-Einstein-condensation-based analogue spacetime”. In: *Phys. Rev. D* 72 (4 Aug. 2005), p. 044020. DOI: [10.1103/PhysRevD.72.044020](https://doi.org/10.1103/PhysRevD.72.044020). URL: <https://link.aps.org/doi/10.1103/PhysRevD.72.044020>.
- [26] Matt Visser and Silke Weinfurtner. *Massive phonon modes from a BEC-based analog model*. 2004. arXiv: [cond-mat/0409639](https://arxiv.org/abs/cond-mat/0409639) [[cond-mat.stat-mech](#)].
- [27] J. M. Vogels et al. “Prediction of Feshbach resonances in collisions of ultracold rubidium atoms”. In: *Phys. Rev. A* 56 (2 Aug. 1997), R1067–R1070. DOI: [10.1103/PhysRevA.56.R1067](https://doi.org/10.1103/PhysRevA.56.R1067). URL: <https://link.aps.org/doi/10.1103/PhysRevA.56.R1067>.

Bibliography XII

- [28] S. Weinfurtner, S. Liberati, and M. Visser. “Analogue Space-time Based on 2-Component Bose-Einstein Condensates”. In: *Quantum Analogues: From Phase Transitions to Black Holes and Cosmology*. Ed. by William G. Unruh and Ralf Schützhold. Berlin, Heidelberg: Springer Berlin Heidelberg, 2007, pp. 115–163. ISBN: 978-3-540-70859-9. DOI: [10.1007/3-540-70859-6_6](https://doi.org/10.1007/3-540-70859-6_6). URL: https://doi.org/10.1007/3-540-70859-6_6.
- [29] Silke Weinfurtner. “Analogue model for an expanding universe”. In: *General Relativity and Gravitation* 37.9 (Sept. 2005), pp. 1549–1554. DOI: [10.1007/s10714-005-0135-7](https://doi.org/10.1007/s10714-005-0135-7). URL: <https://doi.org/10.1007/s10714-005-0135-7>.

Bibliography XIII

- [30] Silke Weinfurtner, Stefano Liberati, and Matt Visser. “Analogue model for quantum gravity phenomenology”. In: *Journal of Physics A: Mathematical and General* 39.21 (May 2006), p. 6807. DOI: [10.1088/0305-4470/39/21/S83](https://doi.org/10.1088/0305-4470/39/21/S83). URL: <https://dx.doi.org/10.1088/0305-4470/39/21/S83>.
- [31] Silke Weinfurtner, Stefano Liberati, and Matt Visser. “Modelling Planck-scale Lorentz violation via analogue models”. In: *Journal of Physics: Conference Series* 33.1 (Mar. 2006), p. 373. DOI: [10.1088/1742-6596/33/1/046](https://doi.org/10.1088/1742-6596/33/1/046). URL: <https://dx.doi.org/10.1088/1742-6596/33/1/046>.